

Alan Xie

360-228-4453 | axie74@gatech.edu | [linkedin.com/in/alanxie/](https://www.linkedin.com/in/alanxie/) | <https://github.com/alanxie>

EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S. Electrical Engineering

Aug. 2025 – Dec. 2028

- GPA: 4.0 — Threads: Telecommunications and Signal Processing
- Relevant Coursework: Linear Algebra, Digital Design, Intro to Signal Processing, Object Oriented Programming

TECHNICAL SKILLS

Languages: Python, C++, Java, SystemVerilog

Software: Git, Altium, Matlab, Vivado, Linux, Docker, Kubernetes

Hardware Skills: Oscilloscope, Logic Analyzer, Soldering, Arduino, Teensy, Spectrum Analyzer

Libraries & Networks: NumPy, SciPy, Matplotlib, Sounddevice, Rtlsdr, RAN, LTE

Clubs & Licenses: Ham Radio Technician License, Amateur Radio Club

EXPERIENCE

Information and Communication Lab Intern

Aug. 2026 – Present

Georgia Tech Research Institute

Atlanta, GA

- Building full scale 5G networks, setting up RUs and configuring CU/DUs to bring up multiple gNBs end to end.
- Working on custom xApp for dedicated radio configuration through a Near RT-RIC, deploying with kubernetes.
- Exploring open source voice communication software and implementing it within RAN.

Speech Synthesis Engineer

May 2026 – Aug. 2026

Neuvat AI

Vancouver, WA

- Designed a voice conversion DSP algorithm in C++ that allows for male to female audio conversions.
- Developed a voice-changing mutation algorithm for arbitrary input wav files, achieving 60% similarity to inputs.
- Aided in speeding up TTS processes, boosting local download speeds.

Vehicle Engineer

Sept. 2025 – May 2026

HyTech Racing @ Georgia Tech

Atlanta, GA

- Worked with 20+ other engineers to create a fully functional electric vehicle with data acquisition.
- Coded 7 digital sensors for temperature data and control of vehicle BMS and other heat-sensitive parts of vehicle.
- Troubleshooted car inverter issues and resolved 50+ lines of C++ firmware, gaining experience in I2C and SPI.

PROJECTS

E2E RF Frontend System

Present

- Using an ESP for IQ data and 4-bit QPSK Modulation, mixing LO signals for analog transmission.
- Implemented a 6-bit word encryption for real-time encoding/decoding, along with custom synch and stop words.
- In the process of designing a transceiver PCB for two-way clocked communication and signal integrity.

Software Defined Radio (SDR) Signal Analysis and Demodulation

Mar. 2026

- Wrote a custom Python script for an RTL-SDR pipeline, capturing and sampling FM signals at 2.4 MS/s.
- Implemented FM demodulation, filtering, and decimation to decode perceivable audio from raw IQ samples.
- Built a complete SDR workflow with real-time decoding and playback for stations between 0-200 MHz range.

DMA-FFT FPGA Spectrum Analyzer

Jun. 2026

- Utilized the Pynq-Z1's hybrid architecture to create a spectrum analyzer, with IP blocks and Python DMA logic.
- Developed a AXI4-Stream pipeline into a Xilinx FFT core, with artificially simulated signals for testing.
- Created a SystemVerilog config module for DMA readiness, validity, and timing checks.